

RING TOPOLOGY

1. Project Overview:

In this section, the topology of focus is the ring topology, in this connection one switch is connected to at least two other switches forming a closed loop all this was implemented using the cisco packet tracer software. All the departments are in their own VLANs with inter-VLAN routes managed by the multilayered switch. Due to the loop STP is enabled to avoid loops even though the design is a loop.

- Project Title: Ring_PathFinder
- Objective: The aim is to demonstrate how data flows in a ring topology and how redundancy can be achieved with closed loops and how they are managed by STP.
- Topology: Ring Topology
- Core Tech: VLAN with segmentation with the system.

2. Materials used:

Item	Quantity	Model
PC	10	PC-PT
Access Switches	5	2960-24TT
Multilayered switch	1	3560-24PS

3. Network design and Topology:

Physical topology:

- Core switch: One multilayered switch handling inter-VLAN
- Distribution switches: 5 access switches connected to the core
- End devices: 10 PCs connected to the distribution switches.

Logical designing with VLAN segmentation that is entirely handled by the IP routing in the multilayered switch.

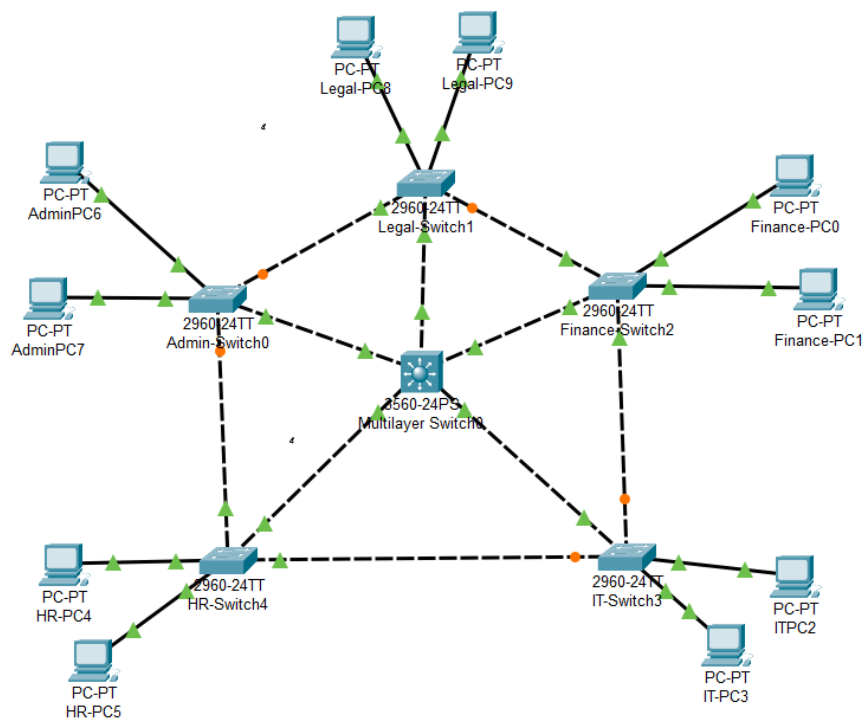


Diagram explanation:

In this set up the access switches are all connected together to form a ring, with each switch having at most two access switches connected to it. It further trunks into the multilayered switch and it is there where the inter-VLAN takes place. Each department has its own VLAN number (10-50), the information can flow five ways at each access switch, it can be from the adjacent switches, the end devices as well as the multicore switch.

However, this design creates a circle, meaning that the connection has a lot of redundancy in it and thus the Spanning Tree Protocol is implemented in the design to ensure that only one port is kept active while the other port is paused. The other paused port will be activated once the currently active one goes down thus ensuring that the information flows correctly and that the system is loop-free. It can be further noticed from the above diagram that every connection between each access switch has one green and another orange, that is the STP at work. The STP ensures that there is still redundancy, but loops are correctly managed.

4. Configurations:

- The CORE SWITCH :

! Basic Configuration

SW-CORE> enable

SW-CORE# configure terminal

SW-CORE(config)# hostname SW-CORE

! Create VLANs

SW-CORE(config)# vlan 10

SW-CORE(config-vlan)# name ADMIN

SW-CORE(config-vlan)# exit

SW-CORE(config)# vlan 20

SW-CORE(config-vlan)# name HR

SW-CORE(config-vlan)# exit

SW-CORE(config)# vlan 30

SW-CORE(config-vlan)# name IT

SW-CORE(config-vlan)# exit

SW-CORE(config)# vlan 40

SW-CORE(config-vlan)# name FINANCE

SW-CORE(config-vlan)# exit

SW-CORE(config)# vlan 50

SW-CORE(config-vlan)# name LEGAL

SW-CORE(config-vlan)# exit

! Configure trunks to access switches
(Gig0/1)

SW-CORE(config)# interface gig0/1

SW-CORE(config-if)# no shutdown

SW-CORE(config-if)# exit

! Configure SVIs for inter-VLAN routing

SW-CORE(config)# interface vlan 10

SW-CORE(config-if)# ip address

192.168.10.1 255.255.255.0

SW-CORE(config-if)# no shutdown

```
SW-CORE(config-if)# exit
```

```
SW-CORE(config)# interface vlan 20
SW-CORE(config-if)# ip address
192.168.20.1 255.255.255.0
SW-CORE(config-if)# no shutdown
SW-CORE(config-if)# exit
```

```
SW-CORE(config)# interface vlan 30
SW-CORE(config-if)# ip address
192.168.30.1 255.255.255.0
SW-CORE(config-if)# no shutdown
SW-CORE(config-if)# exit
```

```
SW-CORE(config)# interface vlan 40
SW-CORE(config-if)# ip address
192.168.40.1 255.255.255.0
SW-CORE(config-if)# no shutdown
SW-CORE(config-if)# exit
```

```
SW-CORE(config)# interface vlan 50
SW-CORE(config-if)# ip address
192.168.50.1 255.255.255.0
SW-CORE(config-if)# no shutdown
SW-CORE(config-if)# exit
```

```
! Enable IP routing
SW-CORE(config)# ip routing
```

```
! Enable Rapid-PVST+ spanning-tree globally
SW-CORE(config)# spanning-tree mode
rapid-pvst
```

```
! Save configuration
SW-CORE(config)# exit
```

```
SW-CORE# copy running-config startup-  
config
```

- ADMIN DEPARTMENT:

```
! Basic Configuration
```

```
SW-ADMIN> enable
```

```
SW-ADMIN# configure terminal
```

```
SW-ADMIN(config)# hostname SW-ADMIN
```

```
! Create VLAN 10
```

```
SW-ADMIN(config)# vlan 10
```

```
SW-ADMIN(config-vlan)# name ADMIN
```

```
SW-ADMIN(config-vlan)# exit
```

```
! Configure PC ports Fa0/3-4
```

```
SW-ADMIN(config)# interface range
```

```
fastEthernet 0/3-4
```

```
SW-ADMIN(config-if-range)# switchport  
mode access
```

```
SW-ADMIN(config-if-range)# switchport  
access vlan 10
```

```
SW-ADMIN(config-if-range)# spanning-tree  
portfast
```

```
SW-ADMIN(config-if-range)# no shutdown
```

```
SW-ADMIN(config-if-range)# exit
```

```
! Configure ring port to next switch Fa0/1
```

```
SW-ADMIN(config)# interface fastEthernet  
0/1
```

```
SW-ADMIN(config-if)# switchport mode  
trunk
```

```
SW-ADMIN(config-if)# switchport trunk  
native vlan 999
```

```
SW-ADMIN(config-if)# no shutdown
```

```
SW-ADMIN(config-if)# exit
```

```
! Configure trunk to MLS Fa0/2
SW-ADMIN(config)# interface fastEthernet
0/2
SW-ADMIN(config-if)# switchport mode
trunk
SW-ADMIN(config-if)# switchport trunk
native vlan 999
SW-ADMIN(config-if)# no shutdown
SW-ADMIN(config-if)# exit
```

```
! Enable Rapid-PVST+ spanning-tree
SW-ADMIN(config)# spanning-tree mode
rapid-pvst
```

```
! Save configuration
SW-ADMIN(config)# exit
SW-ADMIN# copy running-config startup-
config
```

- HR DEPARTMENT:

```
! Basic Configuration
SW-HR> enable
SW-HR# configure terminal
SW-HR(config)# hostname SW-HR
```

```
! Create VLAN 20
SW-HR(config)# vlan 20
SW-HR(config-vlan)# name HR
SW-HR(config-vlan)# exit
```

```
! Configure access ports for PCs (Fa0/3-4)
SW-HR(config)# interface range fastEthernet
0/3-4
SW-HR(config-if-range)# switchport mode
access
```

```
SW-HR(config-if-range)# switchport access  
vlan 20
```

```
SW-HR(config-if-range)# spanning-tree  
portfast
```

```
SW-HR(config-if-range)# no shutdown
```

```
SW-HR(config-if-range)# exit
```

```
! Configure trunk to Core MLS (Fa0/1)
```

```
SW-HR(config)# interface fastEthernet 0/1
```

```
SW-HR(config-if)# switchport mode trunk
```

```
SW-HR(config-if)# switchport trunk native  
vlan 999
```

```
SW-HR(config-if)# no shutdown
```

```
SW-HR(config-if)# exit
```

```
! Enable Rapid-PVST+ spanning-tree
```

```
SW-HR(config)# spanning-tree mode rapid-  
pvst
```

```
! Save configuration
```

```
SW-HR(config)# exit
```

```
SW-HR# copy running-config startup-config
```

- IT DEPARTMENT:

```
! Basic Configuration
```

```
SW-IT> enable
```

```
SW-IT# configure terminal
```

```
SW-IT(config)# hostname SW-IT
```

```
! Create VLAN 30
```

```
SW-IT(config)# vlan 30
```

```
SW-IT(config-vlan)# name IT
```

```
SW-IT(config-vlan)# exit
```

```
! Configure PC ports Fa0/3-4
```

```
SW-IT(config)# interface range fastEthernet  
0/3-4
```

```
SW-IT(config-if-range)# switchport mode  
access
```

```
SW-IT(config-if-range)# switchport access  
vlan 30
```

```
SW-IT(config-if-range)# spanning-tree  
portfast
```

```
SW-IT(config-if-range)# no shutdown
```

```
SW-IT(config-if-range)# exit
```

! Ring ports Fa0/1-2

```
SW-IT(config)# interface range fastEthernet  
0/1-2
```

```
SW-IT(config-if-range)# switchport mode  
trunk
```

```
SW-IT(config-if-range)# switchport trunk  
native vlan 999
```

```
SW-IT(config-if-range)# no shutdown
```

```
SW-IT(config-if-range)# exit
```

! Trunk to MLS Fa0/5

```
SW-IT(config)# interface fastEthernet 0/5
```

```
SW-IT(config-if)# switchport mode trunk
```

```
SW-IT(config-if)# switchport trunk native vlan  
999
```

```
SW-IT(config-if)# no shutdown
```

```
SW-IT(config-if)# exit
```

! Enable Rapid-PVST+ spanning-tree

```
SW-IT(config)# spanning-tree mode rapid-  
pvst
```

! Save configuration

```
SW-IT(config)# exit
```


SW-IT# copy running-config startup-config

- FINANCE DEPARTMENT:

! Basic Configuration

SW-FINANCE> enable

SW-FINANCE# configure terminal

SW-FINANCE(config)# hostname SW-FINANCE

! Create VLAN 40

SW-FINANCE(config)# vlan 40

SW-FINANCE(config-vlan)# name FINANCE

SW-FINANCE(config-vlan)# exit

! Configure PC ports Fa0/3-4

SW-FINANCE(config)# interface range

fastEthernet 0/3-4

SW-FINANCE(config-if-range)# switchport
mode access

SW-FINANCE(config-if-range)# switchport
access vlan 40

SW-FINANCE(config-if-range)# spanning-
tree portfast

SW-FINANCE(config-if-range)# no shutdown

SW-FINANCE(config-if-range)# exit

! Ring ports Fa0/1-2

SW-FINANCE(config)# interface range

fastEthernet 0/1-2

SW-FINANCE(config-if-range)# switchport
mode trunk

SW-FINANCE(config-if-range)# switchport
trunk native vlan 999

SW-FINANCE(config-if-range)# no shutdown

SW-FINANCE(config-if-range)# exit

```
! Trunk to MLS Fa0/5
SW-FINANCE(config)# interface fastEthernet
0/5
SW-FINANCE(config-if)# switchport mode
trunk
SW-FINANCE(config-if)# switchport trunk
native vlan 999
SW-FINANCE(config-if)# no shutdown
SW-FINANCE(config-if)# exit
```

```
! Enable Rapid-PVST+ spanning-tree
SW-FINANCE(config)# spanning-tree mode
rapid-pvst
```

```
! Save configuration
SW-FINANCE(config)# exit
SW-FINANCE# copy running-config startup-
config
```

- LEGAL DEPARTMENT:

```
! Basic Configuration
SW-LEGAL> enable
SW-LEGAL# configure terminal
SW-LEGAL(config)# hostname SW-LEGAL
```

```
! Create VLAN 50
SW-LEGAL(config)# vlan 50
SW-LEGAL(config-vlan)# name LEGAL
SW-LEGAL(config-vlan)# exit
```

```
! Configure PC ports Fa0/3-4
SW-LEGAL(config)# interface range
fastEthernet 0/3-4
```

```
SW-LEGAL(config-if-range)# switchport
mode access
SW-LEGAL(config-if-range)# switchport
access vlan 50
SW-LEGAL(config-if-range)# spanning-tree
portfast
SW-LEGAL(config-if-range)# no shutdown
SW-LEGAL(config-if-range)# exit
```

! Ring ports Fa0/1-2

```
SW-LEGAL(config)# interface range
fastEthernet 0/1-2
SW-LEGAL(config-if-range)# switchport
mode trunk
SW-LEGAL(config-if-range)# switchport trunk
native vlan 999
SW-LEGAL(config-if-range)# no shutdown
SW-LEGAL(config-if-range)# exit
```

! Trunk to MLS Fa0/5

```
SW-LEGAL(config)# interface fastEthernet
0/5
SW-LEGAL(config-if)# switchport mode trunk
SW-LEGAL(config-if)# switchport trunk native
vlan 999
SW-LEGAL(config-if)# no shutdown
SW-LEGAL(config-if)# exit
```

! Enable Rapid-PVST+ spanning-tree

```
SW-LEGAL(config)# spanning-tree mode
rapid-pvst
```

! Save configuration

```
SW-LEGAL(config)# exit
```

SW-LEGAL# copy running-config startup-config

5. IP ADDRESSING:

The network system has an integration of different VLANS assigned to the internal departments. The IPv4 are static.

- VLAN 10 – Admin Department
 - ◆ Network address: 192.168.10.0/24
 - ◆ Subnet Mask: 255.255.255.0
 - ◆ Default gateway: 192.168.10.1
- VLAN 20- HR Department
 - Network Address:192.168.20.0/24
 - Subnet mask:255.255.255.0
 - Default gateway: 192.168.20.1
- VLAN 30- IT Department
 - Network Address: 192.168.30.0/24
 - Subnet mask: 255.255.255.0
 - Default gateway: 192.168.30.1
- VLAN 40- Finance Department
 - Network Address: 192.168.40.0/24
 - Subnet mask: 255.255.255.0
 - Default gateway: 192.168.40.1

- VLAN 50- Legal Department
 - Network Address: 192.168.50.0/24
 - Subnet Mask: 255.255.255.0
 - Default gateway: 192.168.50.1

Device	IP address	VLAN	Remark
Admin-PC1	192.168.10.2	10	Connected to the admin switch

Admin-PC2	192.168.10.3	10	Connected to admin switch
HR-PC1	192.168.20.2	20	Connected to the HR switch
HR-PC2	192.168.20.3	20	Connected to HR switch
IT-PC1	192.168.30.2	30	Connected to IT switch
IT-PC2	192.168.30.3	30	Connected to IT switch
Finance-PC1		40	Connected to the Finance switch
Finance-PC2		40	Connected to the finance switch
Legal-PC1	192.168.50.3	50	Connected to the legal switch
Legal-PC2	192.168.50.2	50	Connected to legal switch

6. Testing and verification:

a) The cdp neighbors as seen by the core switch

```
CoreSwitch#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce  Holdtme    Capability  Platform  Port ID
AdminSwitch      Fas 0/3         133        S           2960      Gig 0/1
HRSwitch         Fas 0/4         133        S           2960      Gig 0/1
FinanceSwitch
                 Fas 0/1         133        S           2960      Gig 0/1
ITSwitch         Fas 0/5         133        S           2960      Gig 0/1
LegalSwitch      Fas 0/2         133        S           2960      Gig 0/1
CoreSwitch#
```

b) A connection can be seen between the core and the access switches, as the core can ping them successfully

```

CoreSwitch#ping 192.168.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/3/4 ms

CoreSwitch#ping 192.168.20.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/5 ms

CoreSwitch#ping 192.168.30.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.30.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/4 ms

CoreSwitch#ping 192.168.40.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.40.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/4 ms

CoreSwitch#ping 192.168.50.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.50.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/8 ms

```

- c) The center switch has an established connection between the end devices and itself

```

CoreSwitch#ping 192.168.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

CoreSwitch#ping 192.168.30.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

CoreSwitch#ping 192.168.40.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.40.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

CoreSwitch#ping 192.168.50.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.50.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

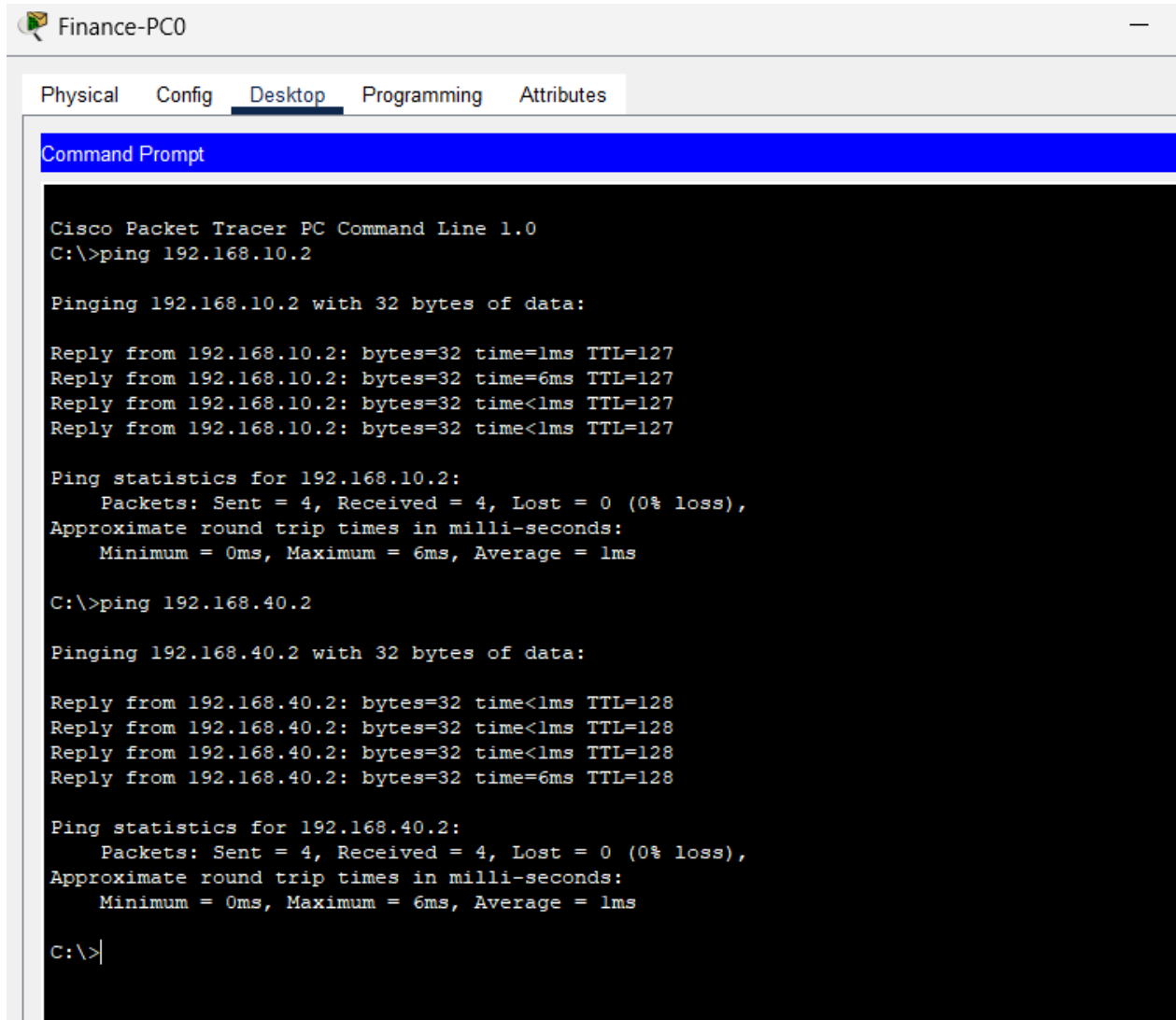
CoreSwitch#ping 192.168.20.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

```

(NOTE: all the outputs appear to be at 80% but that is still okay because the packets still had to discover the path).

- d) Outputs of pinging both internally and external to the VLANs.
(Finance and Administration)



```
Finance-PC0
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=1ms TTL=127
Reply from 192.168.10.2: bytes=32 time=6ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 1ms

C:\>ping 192.168.40.2

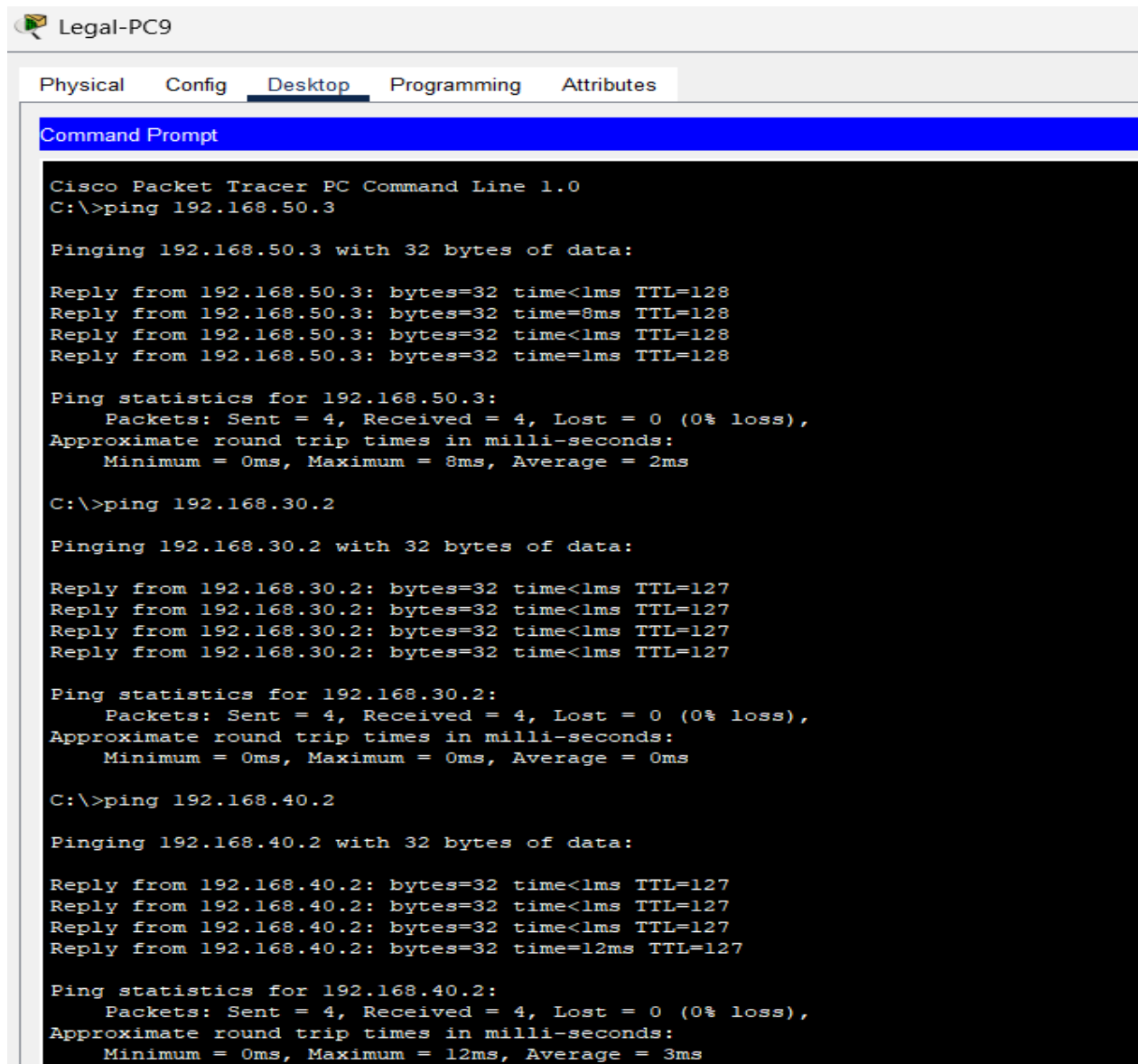
Pinging 192.168.40.2 with 32 bytes of data:

Reply from 192.168.40.2: bytes=32 time<1ms TTL=128
Reply from 192.168.40.2: bytes=32 time<1ms TTL=128
Reply from 192.168.40.2: bytes=32 time<1ms TTL=128
Reply from 192.168.40.2: bytes=32 time=6ms TTL=128

Ping statistics for 192.168.40.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 1ms

C:\>|
```

(Communication between Legal, IT and Finance)



The screenshot shows the Cisco Packet Tracer PC Command Line interface for a device named 'Legal-PC9'. The 'Desktop' tab is selected. The command prompt shows three ping commands being executed: 'ping 192.168.50.3', 'ping 192.168.30.2', and 'ping 192.168.40.2'. Each command is followed by a detailed response showing the ping statistics, including the number of packets sent and received, the loss percentage, and the approximate round trip times in milliseconds.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.50.3

Pinging 192.168.50.3 with 32 bytes of data:

Reply from 192.168.50.3: bytes=32 time<1ms TTL=128
Reply from 192.168.50.3: bytes=32 time=8ms TTL=128
Reply from 192.168.50.3: bytes=32 time<1ms TTL=128
Reply from 192.168.50.3: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.50.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 2ms

C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.40.2

Pinging 192.168.40.2 with 32 bytes of data:

Reply from 192.168.40.2: bytes=32 time<1ms TTL=127
Reply from 192.168.40.2: bytes=32 time<1ms TTL=127
Reply from 192.168.40.2: bytes=32 time<1ms TTL=127
Reply from 192.168.40.2: bytes=32 time=12ms TTL=127

Ping statistics for 192.168.40.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms
```

Based on these two replies, one can safely say that Inter-VLAN is working as it should.

7. Advantages:

- Due to the loop that the diagram has created, whenever there is a point of failure the information simply changes its route.
- VLANs are simply and single handedly handled by the MLS
- There is less congestion at the core as traffic is isolated by the different VLANs.
- Expanding the network is as easy as connecting to the core then trunk it

8. Disadvantages:

- The topology requires extensive and careful configuration; misconfigurations lead to blocked links even loops.
- The need for cables rises very much as a single device may need more than 2 cables.
- The information is required to pass through a lot of paths; thus latency can increase especially if it finds a link closed.

9. Conclusion:

It can now be concluded that the ring topology provides a balanced approach between redundancy and segmentation. The network ensures that the devices within the same network and even outside can communicate efficiently with the multilayered switch managing the inter-VLAN routing. The rapid-pvst and spanning trees ensure that the loops in the system can be properly treated. Despite the redundancy, the core continues to be a critical zone in the setup because if it's down, there will be no communication in the network, this highlights how risky it is to place one device as a core of dependency for all the other devices in the network.

Overall, the ring topology offers both reliability and efficiency while ensuring that VLAN segmentation is properly met. It is best used in places where redundancy is essential in keeping the organization going, provided that the core is taken care of. Proper configuration leads to a secure, reliable, robust solution to organization in their network systems.